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Sentiment analysis on user reviews

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Abstract

We study five-class sentiment prediction for bilingual English-German review text, evaluated by the score $1 - \text{MAE}/4$. We compare BoW and TF-IDF baselines with XLM-RoBERTa fine-tuning, parameter-efficient LoRA, ordinal and distributional losses, cross-lingual consistency, punctuation features, per-language routing, and probability averaging. XLM-R-large with LoRA provides the main improvement over lexical baselines, while the best validation score, 0.9102, comes from an ensemble of XLM variants whose public Kaggle score rounds to 0.909, tied with our best single backbone once its decoding rule is calibrated on validation. Our ablations suggest that most gains come from multilingual pretraining and scale, with punctuation features, language-specific routing, and ensembling providing small but measurable improvements, while ordinal objectives are not consistently beneficial in our single-seed setting.

1. Introduction

We study bilingual five-class sentiment prediction for English and German product reviews. The labels are ordinal stars $\{0, 1, 2, 3, 4\}$ and the competition score is $1 - \text{MAE}/4$, so the distance between predicted and true labels matters. The training set contains 252,000 labelled reviews and is balanced both by class (50,400 reviews per star) and by language after fasttext language identification; the hidden test set has 168,000 reviews with the same English/German balance. Review lengths are moderate (median 168 characters, 95th percentile 564), so we use a 256-token cap and no language-specific cleaning beyond the tokenizer.

Our experiments compare classical TF-IDF/bag-of-words baselines with XLM-RoBERTa fine-tuning, LoRA, ordinal losses, per-language routing, cross-lingual consistency,

regression-style decoding, punctuation features, and ensembling. The main question is which additions improve MAE rather than only categorical accuracy.

Contributions. We benchmark twelve classical baselines to establish a reproducible floor; compare full fine-tuning against parameter-efficient LoRA on XLM-RoBERTa; test whether ordinal-aware objectives and regression-style decoding help the MAE metric; route English and German reviews through monolingual backbones; add a cross-lingual consistency objective based on back-translation; and inspect a tiny punctuation-feature head for interpretable typographic cues.

2. Models and Methods

Baselines. We first establish a classical floor with bag-of-words and TF-IDF features, each using unigrams and bigrams. On top we train logistic regression (Chung, 2020), a linear SVC, ridge regression (Hilt & Seegrist, 1977), an SGD log-loss classifier, a multilayer perceptron, and LightGBM.

XLM-RoBERTa with LoRA. Our main backbone is XLM-RoBERTa (Conneau et al., 2020). For XLM-R-large we avoid updating all $\approx 0.6\text{B}$ parameters by applying LoRA (Hu et al., 2021) to the attention query and value projections:

$$h = Wx + \frac{\alpha}{r}BAx, \quad r = 16, \quad \alpha = 32.$$

With dropout 0.05, this leaves only $\approx 0.16\%$ of the backbone trainable. The classification head is a linear projection from the pooled $\langle s \rangle$ representation to the five star classes.

Ordinal losses. Because the metric is MAE-based, we compare naive cross-entropy with two ordinal-aware soft targets. Neighbor smoothing, an ordinal variant of label smoothing (Szegedy et al., 2016), moves mass $\varepsilon = 0.1$ from the true class to its adjacent labels, folding edge mass back into classes 0 and 4. A distance-aware target instead assigns probability proportional to $\exp(-|k - y|/\tau)$ with $\tau = 1.0$. Both keep near misses closer to the target than far-away errors. For logits z , standard CE optimises

$$\mathcal{L}_{\text{CE}}(y, z) = -\log p_y, \quad p_k = \frac{\exp z_k}{\sum_{j=0}^4 \exp z_j},$$

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which treats predicting 1 and 4 for a true class 0 as equally wrong. The smoothed objectives replace the one-hot target with a distribution $q(y)$ and minimise $-\sum_k q_k(y) \log p_k$.

Language routing and consistency. For the per-language pipeline we route examples with fasttext `lid.176` (Joulin et al., 2017): English reviews use `roberta-large` (Liu et al., 2019), German reviews use `deepset/gbert-large` (Chan et al., 2020), and both adapters use the same LoRA configuration as XLM-R-large. We also test a cross-lingual consistency variant: each example is back-translated with MarianMT (Junczys-Dowmunt et al., 2018), and training adds a symmetric KL penalty between original and translated predictions with weight $\lambda = 0.3$.

Punctuation and ordinal decoding. To expose review-specific intensity cues, we add a 20-parameter head over $\log(1 + n)$ counts of exclamation marks, question marks, ellipses, and all-caps tokens:

$$\text{logits}(x) = \text{XLM-R}_{\text{LoRA}}(x) + \phi(x)^\top W, \quad W \in \mathbb{R}^{4 \times 5}.$$

We also train a regression head with MSE on scalar star values; its output is discretised by validation-tuned monotone thresholds $\tau_0 < \tau_1 < \tau_2 < \tau_3$. For the cross-lingual model, probabilities from the original and translated test views are averaged before threshold decoding. We tune temperature (Guo et al., 2017) and thresholds on validation because the leaderboard metric rewards calibrated near misses more than raw gmax accuracy.

Strengthened punctuation model. Our strongest single-backbone configuration (PUNCT-V2) keeps the $\phi(x)^\top W$ head but tightens the rest of the pipeline. LoRA is widened to $r=32$, $\alpha=64$ and applied to the query, key, value, and output projections; the learning rate is reduced to 2×10^{-4} . Inputs are normalised before tokenization (emoji demojization, collapsing runs of ≥ 4 identical characters to three) while punctuation features are still extracted from the *raw* text so that intensity cues are preserved. At test time we fit a single temperature on the validation NLL (Guo et al., 2017) and search the four monotone thresholds $\tau_0 < \dots < \tau_3$ that maximise the $1 - \text{MAE}/4$ score with Nelder-Mead (Nelder & Mead, 1965), applied to the calibrated expected-value scores $\sum_k k p_k(x)$.

Optimisation. All deep models use a 90/10 stratified train/validation split (`random_state=42`), AdamW with weight decay 0.01, linear warmup over 6% of steps, fp16, and three epochs. LoRA runs use learning rate 3×10^{-4} and batch size 32 except the cross-lingual run (batch 16); full fine-tuning uses 2×10^{-5} and batch 16.

Table 1. Effect of the training loss on XLM-R-large + LoRA, single seed.

Loss	Acc	MAE	Score
Naive CE	0.6698	0.3673	0.9082
Neighbor-smoothed	0.6702	0.3664	0.9084
Distance-aware	0.6688	0.3682	0.9080

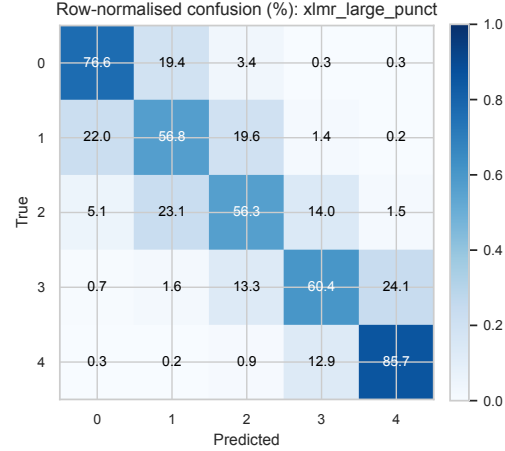


Figure 1. Row-normalised confusion matrix (%) of XLM-R-large + LoRA + punct. Errors concentrate on adjacent classes; extreme confusions are nearly absent.

3. Results

All numbers use the same 25,200-example validation split. The three training losses on XLM-R-large + LoRA are effectively tied: naive CE scores 0.9082, neighbor-smoothed CE 0.9084, and the distance-aware target 0.9080. We therefore keep neighbor smoothing for the remaining classification models because it matches the MAE metric without hurting performance.

Table 2 shows that pretrained multilingual encoders dominate the lexical baselines: the weakest deep run, XLM-R-base + LoRA, is already +2.65 score points above TF-IDF logistic regression. XLM-R-large LoRA matches full fine-tuning while training $\approx 0.16\%$ of the backbone and using about half the time. The single-backbone variants span 0.9073–0.9087 under the default recipe, and per-language routing reaches 0.9097. Our strengthened PUNCT-V2 configuration posts 0.9101 on validation, but this number is partly in-sample: its temperature and four decoding thresholds are fit on the validation split itself (the same caveat applies to the regression row). The held-out comparison on the public Kaggle split (see §4) is therefore the fair one, where PUNCT-V2 ties the nine-model ensemble at 0.909.

Figure 1 explains why ordinal modelling matters but also why it does not move the score much here: most residual

Table 2. Validation results. Score = $1 - \text{MAE}/4$; train time is wall-clock seconds. Classical rows show the strongest or representative baselines from the full sweep.

Model	Acc	MAE	Score	Time (s)
BoW + Ridge	0.4840	0.7368	0.8158	158
BoW + LinSVC	0.5183	0.6362	0.8409	105
TF-IDF + MLP	0.5329	0.5990	0.8502	4200
TF-IDF + SGD	0.5681	0.5838	0.8541	0.46
BoW + MLP	0.5379	0.5852	0.8537	4194
TF-IDF + LinSVC	0.5652	0.5481	0.8630	8.8
TF-IDF + Ridge	0.5712	0.5428	0.8643	7.8
BoW + LogReg	0.5680	0.5321	0.8670	287
BoW + LGBM	0.5850	0.5238	0.8690	97
TF-IDF + LGBM	0.5858	0.5235	0.8691	255
BoW + SGD	0.5932	0.5037	0.8741	1.2
TF-IDF + LogReg	0.5956	0.4972	0.8757	5.6
XLM-R-base, LoRA	0.6543	0.3912	0.9022	529
XLM-R-large, full FT	0.6642	0.3725	0.9069	2426
XLM-R-large, LoRA + xling	0.6662	0.3710	0.9073	2852
XLM-R-large, LoRA	0.6702	0.3664	0.9084	1340
XLM-R-large, LoRA + reg	0.6630	0.3655	0.9086	1347
XLM-R-large, LoRA + punct	0.6715	0.3650	0.9087	1324
Per-language LoRA	0.6736	0.3611	0.9097	1286
XLM-R-large, LoRA + punct-v2	0.6691	0.3594	0.9101	4149
Ensemble (XLM variants)	0.6718	0.3594	0.9102	—

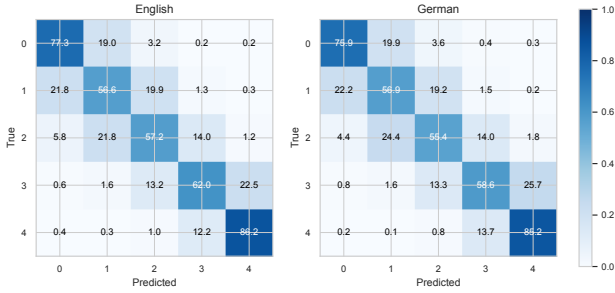


Figure 2. Same model as Figure 1, restricted to English (left) and German (right) reviews. German errors spread further off-diagonal, especially around classes 1–3.

Table 3. Validation score split by detected language.

Model	En	De
XLM-R-large, LoRA	0.9098	0.9070
Per-language LoRA	0.9122	0.9073

mistakes are already off-by-one errors around the middle classes. The model almost never confuses class 0 with 4.

The per-language gain is mostly English-driven: roberta-large raises the English score from 0.9098 to 0.9122, while deepset/gbert-large only nudges German from 0.9070 to 0.9073. This suggests XLM-R already transfers well to German on the balanced bilingual corpus, whereas English benefits more from

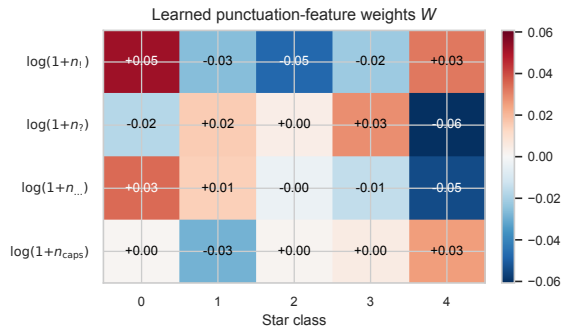


Figure 3. Learned additive weights W of the punctuation head. Rows are log-counted features; columns are star classes.

a specialised monolingual encoder. The language-split confusion matrices support the same conclusion. German reviews spread more probability mass into neighbouring classes around 1–3, while English is sharper on the diagonal. However, neither language shows many extreme mistakes, so the per-language pipeline mainly reduces ordinary boundary errors rather than fixing a qualitatively different failure mode.

The punctuation head is small enough to inspect directly (Figure 3). Exclamation marks push both extreme classes (0 and 4) upward and suppress the middle, consistent with intensity rather than polarity. Ellipses raise low-star logits and suppress class 4, question marks mainly lower the five-star logit, and all-caps has the weakest effect. All weights

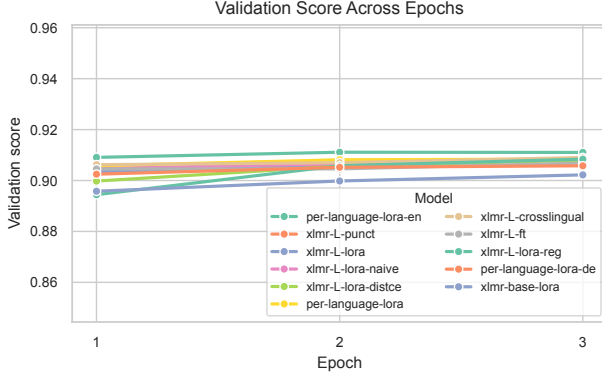


Figure 4. Validation score per epoch for the strongest deep configurations.

stay below 0.061, so the head acts as lightweight calibration rather than a standalone sentiment cue.

Training dynamics. Figure 4 shows validation score by epoch for the strongest deep configurations. All XLM variants continue improving through epoch 3, but the gap between deep models is much smaller than the jump from classical baselines to XLM-R. We therefore kept the schedule fixed rather than spending compute on longer runs whose expected gain was below the single-seed uncertainty.

4. Discussion

LoRA vs full fine-tuning. LoRA at $r=16$ on XLM-R-large reaches 0.9084 against 0.9069 for full fine-tuning, while training only $\approx 0.16\%$ of the backbone in roughly half the time. On a task of this size, full fine-tuning of a 0.6B-parameter encoder buys nothing, and the cheap LoRA runs are what make the rest of the ablation grid affordable.

Per-language vs multilingual. Per-language routing improves validation from 0.9084 to 0.9097, driven almost entirely by English (*roberta-large*: 0.9098 $\rightarrow$ 0.9122); *deepset/gbert-large* only nudges German from 0.9070 to 0.9073. This contradicts our original hypothesis: we expected the German monolingual model to be the main lift, but XLM-R is already strong on German and *roberta-large* is the cleaner upgrade on English.

Cross-lingual consistency and decoding. The symmetric-KL consistency term scores 0.9073, slightly under plain LoRA at roughly double the cost. The MSE regression head with tuned thresholds (0.9086) ties the best classifier, so once the backbone has learned the class ordering, swapping the ordinal decoding rule mostly reallocates probability among adjacent classes.

Punctuation and ensembling. Adding four punctuation features gives the best classifier under the default LoRA recipe (0.9087); learned weights mostly encode intensity. PUNCT-V2 presents wider LoRA across q/k/v/out, lower learning rate, input normalisation, and post-hoc temperature plus threshold tuning; the model posts 0.9101 on validation, but its temperature and four thresholds are fit on that same split, so the number is partially in-sample. The held-out comparison comes from the public Kaggle split: PUNCT-V2 and the ensemble tie at 0.909, per-language LoRA follows at 0.908, the remaining XLM-large variants cluster at 0.907, full fine-tuning at 0.906, and XLM-R-base at 0.902. The ordering matches validation (evidence the Table 2 deltas are not pure overfitting), but the apparent jump from 0.9087 to 0.9101 collapses to about 0.002 once measured fairly: a careful single model is competitive with averaging once decoding is aligned with the metric, not strictly better. The ensemble still helps MAE more than accuracy, meaning it corrects severity rather than adding exact matches. Final grading is on a private split we cannot observe.

Limitations. We run a single random seed per configuration, so differences smaller than ≈ 0.001 should be read as noise. We did not try explicit ordinal-regression heads (CORAL / CORN (Cao et al., 2020)), confidence-gated consistency training, or longer schedules, any of which could plausibly close the small remaining gap. Most reported numbers are on the held-out validation split; on the test set we observe only the public-leaderboard score and grading happens on a separate, unobserved private split.

5. Summary

XLM-R-large with LoRA provides the main jump over lexical baselines. Per-language routing reaches 0.9097 validation (mostly via English), and a nine-model ensemble is best at 0.9102 validation / 0.909 public Kaggle. A single careful backbone with calibrated decoding (PUNCT-V2) ties the ensemble on Kaggle, although its validation score is partially in-sample because the decoding rule is tuned on that split. The central lesson is that multilingual pretraining and parameter-efficient adaptation matter far more than the exact ordinal objective on this balanced bilingual dataset: ordinal losses, thresholded regression, and the punctuation features are all defensible, but their differences are smaller than the single-seed uncertainty. A reliable recipe is therefore: start from XLM-R-large, adapt with LoRA, calibrate decoding for the MAE metric, and only ensemble after individual models have been validated.

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